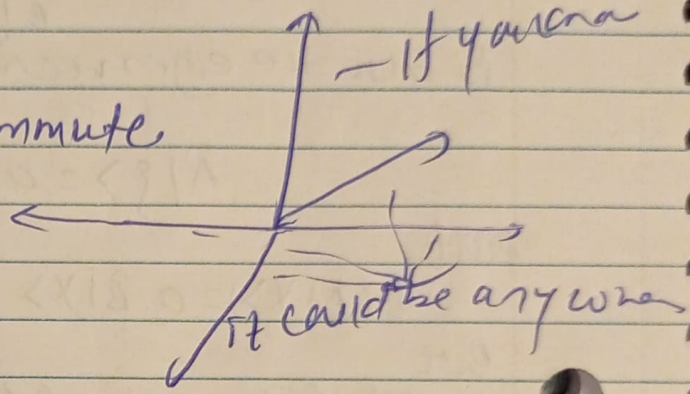
we can't distinguish the vectors $|\alpha_1\rangle$, $|\alpha_2\rangle$ & $|\alpha\rangle$, we can't really decide the dirⁿ.

(Parity is only when real space representation, we can use to distinguish)

You can't have one projection but other could be anywhere

If the operator don't commute,



Instead of using parity we can think of

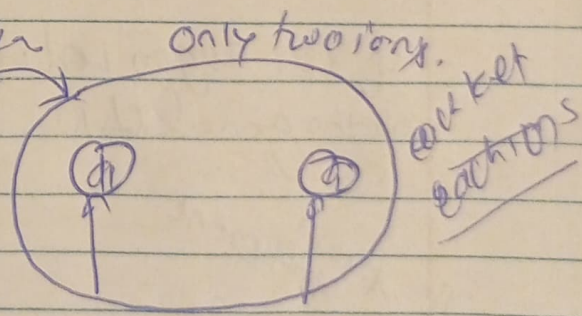
$\hat{A}|b_1\rangle = b_1|\alpha_1\rangle$

$\hat{A}|b_2\rangle = b_2|\alpha_2\rangle$

If they commute or not.

If they are CSCO, they have common basis,

Q4 Set of hilbert space for orbital $\{|\phi_1\rangle, |\phi_2\rangle\}$ ≙ generate my $\mathcal{H}$



identical same energy required to put two if it lands on $|\phi_1\rangle$ two if " " $|\phi_2\rangle$ } it is the energy required to put electron (which is potential) (generacy).

$$\mathcal{H}_{pc} = \begin{pmatrix} |\phi_1\rangle & |\phi_2\rangle \\ \text{two} & 0 \\ 0 & \text{two} \end{pmatrix}$$ matrix

$\hat{H}_{PE} = \hbar\omega_0 |\phi_1\rangle\langle\phi_1| + \hbar\omega_0 |\phi_2\rangle\langle\phi_2|$

we can write it as sum as projectors, (spectral decomposite).

(Same representation)

$= \hbar\omega_0 (|\phi_1\rangle\langle\phi_1| + |\phi_2\rangle\langle\phi_2|)$

$\hat{\mathbb{1}}$ identity help of completeness.

$\therefore \hat{H}_{PE} = \hbar\omega \hat{\mathbb{1}}$ potential energy identity in this a

$\hat{H}_{PE} = \hbar\omega \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

But electron can go in other state (we can't control),

$\therefore KE \neq |\phi_1\rangle \longrightarrow |\phi_2\rangle$ with intensity Δ.

$|\phi_2\rangle \longrightarrow |\phi_1\rangle$

$\hat{H}_{KE} = \begin{pmatrix} |\phi_1\rangle & |\phi_2\rangle \\ 0 & \Delta \\ \Delta & 0 \end{pmatrix}$

$\hat{H}|\phi_1\rangle + \hat{H}|\phi_2\rangle = \Delta|\phi_2\rangle + \Delta|\phi_1\rangle$.

$\Delta|\psi_1\rangle\langle\psi_2| + \Delta|\psi_2\rangle\langle\psi_1|$

when we combine.

Now, $E = PE + KE$.

$\hat{H} = \hat{H}_{PE} + \hat{H}_{KE}$.

$\hat{H} = \begin{pmatrix} |\phi_1\rangle & |\phi_2\rangle \\ \hbar\omega_0 & \Delta \\ \Delta & \hbar\omega_0 \end{pmatrix}$

check for commute → If commute LECT of $\hat{H}_{PE}$ is identity & anything commute with identity.

$[\hat{H}_{PE}, \hat{H}_{KE}] = 0$ → they share common basis.

To know where electron land exactly, we diagonalize $(\hat{H})$.

$$\det \begin{vmatrix} \hbar\omega_0 - \lambda & \Delta \\ \Delta & \hbar\omega_0 - \lambda \end{vmatrix} = 0$$

$$(\hbar\omega_0 - \lambda)^2 - \Delta^2 = 0$$

$$\hbar\omega_0 - \lambda^2 = \Delta^2$$

$$\lambda_1 = \hbar\omega_0 + \Delta$$

$$\lambda_2 = \hbar\omega_0 - \Delta$$

If we measure total energy then we get either

$$\lambda_1 \text{ or } \lambda_2$$

Here, we know electron will go in which configuration

Now, we need to find eigenvector;

$$\lambda = \lambda_1$$

$$\begin{pmatrix} \hbar\omega_0 & \Delta \\ \Delta & \hbar\omega_0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (\hbar\omega_0 + \Delta) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\hbar\omega_0 x + \Delta y = \hbar\omega_0 x + \Delta x$$

$$\Delta y = \Delta x$$

$$y = x.$$

Normalize

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\phi_1\rangle + |\phi_2\rangle)$$

for, $\lambda = -\lambda_2$.

$$\begin{pmatrix} \hbar\omega_0 & \Delta \\ \Delta & \hbar\omega_0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = (\hbar\omega_0 - \Delta)\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\hbar\omega_0 x + \Delta y = \hbar\omega_0 x - \Delta x$$
$$\Delta y = -\Delta x$$
$$y = -x.$$

$$|\vec{r}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|\vec{r}_2\rangle = \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_2\rangle).$$

now, we need to go position representation.

① $\hat{H}_{PE}$ & $\hat{H}_{ICE}$ are CSCO
2) they commute $[\hat{H}_{PE}, \hat{H}_{CC}] = 0$

② Diffeomtrate,

	$\hat{H}_{PE}$	$\hat{H} = \hat{H}_{PE} + \hat{H}_{ICE}$
$\|r_1\rangle$	$= \hbar\omega_0$	$\hbar\omega_0 + \Delta$
$\|r_2\rangle$	$= \hbar\omega_0$	$\hbar\omega_0 - \Delta$.

To get a "feely" go to $\{|\vec{r}\rangle\}$.

$|\phi_1\rangle \rightarrow \langle\vec{r}|\phi_1\rangle = \phi_1(\vec{r})$
$|\phi_2\rangle \rightarrow \langle\vec{r}|\phi_2\rangle = \phi_2(\vec{r})$

when we measure probability we the number.

$d\rho = ?$, $d\rho = |\langle r|r_1\rangle|^2 d^3r$

← from eigen value equation

$$d\rho = |\langle \vec{r} | \left(\frac{1}{\sqrt{2}}|\phi_1\rangle + |\phi_2\rangle\right)|^2 d^3r$$

$$= \left| \frac{\langle \vec{r}|\phi_1\rangle}{\sqrt{2}} + \frac{\langle \vec{r}|\phi_2\rangle}{\sqrt{2}} \right|^2 d^3r$$

$$= \left| \frac{\phi_1(r)}{2} + \frac{\phi_2(r)}{2} \right|^2 d^3r$$

$$= \frac{d^3r}{2} \left[(\phi_1^*(\vec{r}) + \phi_2^*(\vec{r}))\right]\left[\phi_1(\vec{r}) + \phi_2(\vec{r})\right]$$

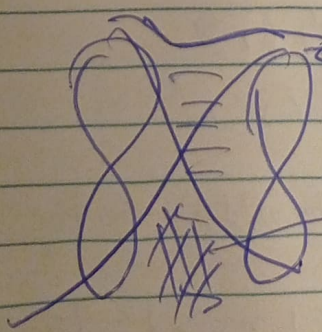
$$d\rho(|\vec{r}\rangle) = \frac{d^3r}{2}\left(\phi_1^*\phi_1 + \phi_2^*\phi_2 + \underbrace{\phi_1^*\phi_2 + \phi_2^*\phi_1}_{2\,Re\,\phi_2^*\phi_1}\right)$$

$$d\rho(|\vec{r_2}\rangle) = \frac{d^3r}{2}\left(\quad\right)$$

Short cut

$$\langle \vec{r} | \frac{1}{\sqrt{2}}(|\phi_1\rangle - |\phi_2\rangle)|^2 d^3r$$

$$= \frac{d^3r}{2}\left(\phi_1^*\phi_1 + \phi_2^*\phi_2 - \underbrace{\phi_1^*\phi_2 - \phi_2^*\phi_1}_{-2\,Re\,\phi_2^*\phi_1}\right)$$



bonding and antibonding.